

Question Paper

Exam Date & Time: 26-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: PHY 1003/PHY_1003 APPLIED PHYSICS FOR ENGINEERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain with a suitable diagram the construction and working principle of a semiconductor laser. (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) An optic fibre of core refractive index 1.80 is to be cladded to ensure that the refracted light ray grazes the interface. The optical fibre will contain light travelling within 10° of the fibre axis when it is placed in air medium. What minimum refractive index is allowed for the cladding? If the same optical fiber is placed in water medium (refractive index of water = 1.33), then find out the acceptance angle and numerical aperture. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) What is modal dispersion? Which type of optical fibre can be used to remove this kind of dispersion? (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) Define (i) group speed and (ii) phase speed. Show that for a given non-relativistic quantum particle, the group speed of a wave packet is equal to the particle speed. (5)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) When a free electron scatters a 0.854 nm x-ray photon, the electron recoils with a speed of 1.48×10^6 m/s. What is the resulting Compton shift in the photon's wavelength? (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) V_1 and V_2 are the stopping potentials for the light of two different frequencies ν_1 and ν_2 ($\nu_2 > \nu_1 > \nu_0$) respectively, when it is incident on a photo emissive surface. Is the relation $V_1 < V_2$ TRUE or FALSE. Justify your answer. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) Obtain the general solution to the Schrödinger equation for a particle of mass m in a finite potential well of height U and length L . Mention the boundary conditions to solve this system. Sketch the probability density for the lowest two energy states. (5)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) A student conducted a photoelectric effect experiment by shining light of various wavelengths onto a metal surface. To stop the flow of photocurrent, the student applied a negative potential until the current dropped to zero. The following readings were recorded. Are these readings valid? Provide justification for your answer. (3)

Wavelength (nm)	Stopping Potential (V)
400	1.71
450	1.08
500	0.86
550	1.36

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 3C) **Increasing the mass of a particle enhances its tunneling probability for a fixed barrier** State whether the given statement is True or False. Justify your answer. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Explain the phenomenon of single-particle interference with reference to the double-slit experiment. Illustrate your answer with suitable diagrams and discuss how this experiment demonstrates the probabilistic nature of quantum mechanics and the principle of superposition. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) What are Bell states, and how do they illustrate the concept of quantum entanglement? (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) **The S gate changes the probability of measuring a qubit in the computational basis.** State whether the given statement is True or False. Justify your answer. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) Explain the working principles of the CNOT and CCNOT gates with the help of circuit diagrams and truth tables. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Prove the following identities:
a) $-\otimes - = -$ b) $-\otimes - = -$ c) $-\otimes - = -$ (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) **The H gate leaves the $|+\rangle$ state unchanged**. State whether the given statement is True or False. Justify your answer. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_1003:APPLIED PHYSICS FOR ENGINEERS

#	Course Outcome	Description	Marks
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1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	7
4	4	CO 4	20
5	5	CO 5	3
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

PHY_1003:APPLIED PHYSICS FOR ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	4	20(40%)
3	Application	7	21(42%)
4	Analysis	4	9(18%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)